

FLOODRIX

Predict. Protect. Deliver.

PRODUCT REQUIREMENTS DOCUMENT

AI-Powered Flood Prediction & Logistics Intelligence Platform

IBM × Kerala Government Hackathon 2026

Challenge 5 — Flood-Aware Logistics Planner for Kerala

Version: 1.0

Core ML Features: Latitude, Longitude, Rainfall_1h, Rainfall_6h, Rainfall_24h, Elevation, Historical_flood_frequency, Flood_zone

Target Variable: Flood_occurred

1. Executive Summary

FLOODRIX is an AI-powered flood prediction and logistics intelligence platform designed to predict the probability of flooding at specific geographic locations and along active logistics routes.

The core of FLOODRIX is a machine-learning model trained using historical flood occurrence data combined with geographical, rainfall, elevation, and historical flood-frequency information. The model learns the relationship between environmental conditions and actual historical flood occurrences.

The model uses eight input features — Latitude, Longitude, Rainfall_1h, Rainfall_6h, Rainfall_24h, Elevation, Historical_flood_frequency, and Flood_zone — with Flood_occurred as the target variable.

During prediction, current and forecast rainfall conditions are combined with static geographic and historical flood characteristics to estimate the probability of flooding at locations along a logistics route.

The resulting flood probability is then integrated with route information, vehicle ETA, cargo type, delivery deadlines, and road-closure information to recommend safer and more reliable logistics routes.

2. Problem Statement

Kerala experiences recurring monsoon-related disruptions including flooding, waterlogging, landslides, and road closures. These disruptions can cause delivery delays, increased fuel consumption, increased travel time, cargo losses, spoilage of perishable goods, and driver safety risks.

Small logistics operators may become aware of a disruption only after a vehicle has already started its journey.

Traditional navigation systems primarily focus on distance, traffic, and travel time. FLOODRIX addresses the additional question: **What is the probability that this route will be affected by flooding when the vehicle reaches it?**

The official Challenge 5 calls for route visualization, weather alerts, flood-risk mapping, alternate route suggestions, and delivery ETA estimation.

3. Product Vision

To transform historical flood information into predictive logistics intelligence by estimating the probability of flooding along a vehicle's route and using that prediction to support safer routing decisions.

4. Product Objectives

- Learn flood patterns from historical data.
- Predict flood occurrence probability for a geographic location.
- Apply the prediction to road segments along a logistics route.
- Consider the expected arrival time of the vehicle.
- Identify potentially high-risk road segments.
- Combine flood risk with road closures and logistics constraints.
- Recommend safer alternative routes.
- Continuously update route risk during an active journey.

5. Core Concept

Layer 1 — Flood Prediction

Latitude
Longitude
Rainfall_1h
Rainfall_6h
Rainfall_24h
Elevation
Historical_flood_frequency
Flood_zone
↓
ML Model
↓
Flood Probability

Layer 2 — Logistics Decision

Flood Probability
+
Route
+
Vehicle ETA
+
Cargo
+
Delivery Deadline
+
Road Closure
↓
Logistics Decision Engine
↓
GO / REROUTE / WAIT

The ML model answers: **What is the probability of flooding here?** The logistics engine answers: **Given that risk, what should the logistics operator do?**

6. Target Users

User	Primary Needs
Logistics Operators	Plan shipments, compare routes, monitor risk, receive alerts, make rerouting decisions
Lorry / Truck Operators	Check route conditions, receive flood alerts, identify risky segments, follow alternatives
FMCG Distributors	Protect delivery schedules and monitor disruption
Agricultural Traders	Protect perishable cargo and reduce spoilage

7. ML Prediction Model

7.1 Exact Input Features

Feature	Description
Latitude	Geographic latitude of the observation
Longitude	Geographic longitude of the observation
Rainfall_1h	Rainfall accumulated during the previous 1 hour
Rainfall_6h	Rainfall accumulated during the previous 6 hours
Rainfall_24h	Rainfall accumulated during the previous 24 hours
Elevation	Elevation of the location
Historical_flood_frequency	Frequency of historical flood occurrence at or around the location
Flood_zone	Historical flood-zone classification associated with the location

7.2 Target Variable

Flood_occurred is the target variable.

0 → Flood did not occur
1 → Flood occurred

The initial ML problem is a binary classification problem: given environmental and geographic conditions, predict the probability that flooding will occur at a particular location.

8. Training Dataset

Each training record represents a geographic location under a particular set of environmental conditions.

```
Latitude, Longitude, Rainfall_1h, Rainfall_6h, Rainfall_24h, Elevation, Historical_flood_frequency, Flood_zone, Flood_occurred
10.52, 76.21, 32, 98, 184, 18, 0.72, 1, 1
10.54, 76.23, 8, 22, 47, 31, 0.24, 0, 0
10.51, 76.25, 41, 126, 230, 12, 0.81, 1, 1
```

The model learns patterns connecting rainfall, location, elevation, historical flood frequency, and flood zone to flood occurrence.

9. Feature Definitions

Latitude / Longitude

Provide geographic context and allow the model to learn spatial relationships.

Rainfall_1h

Captures short-term rainfall intensity and sudden heavy rainfall events.

Rainfall_6h

Captures medium-term accumulated rainfall and sustained rainfall conditions.

Rainfall_24h

Captures long-duration accumulated rainfall.

Elevation

Represents the geographic elevation of the location.

Historical_flood_frequency

Represents how frequently flooding has historically occurred at or around the location.

Flood_zone

Represents the historical flood-zone classification associated with the location.

Flood_occurred

Observed outcome: 0 for no flood and 1 for flood.

10. Training Pipeline

```
Historical Flood Data
+
Historical Rainfall
+
Elevation Data
+
```

```
Flood Zone Data
↓
Data Cleaning
↓
Geospatial Matching
↓
Feature Engineering
↓
Training Dataset
↓
Train / Validation / Test Split
↓
Model Training
↓
Model Evaluation
↓
Trained Model
```

11. Prediction Pipeline

```
Current Location
+
Current / Forecast Rainfall
+
Elevation
+
Historical Flood Frequency
+
Flood Zone
↓
Feature Vector
↓
Trained ML Model
↓
Flood Probability
```

The same model input schema must be maintained at inference time. Historical outcome information such as actual future flood occurrence is never used as an input to a future prediction.

12. Prediction Along a Route

A route is not treated as one geographic point. The route is divided into road segments or sampled geographic points. Each segment is assigned the required model features and passed through the trained model.

Segment	Flood Probability	Risk
Segment 1	12%	Low
Segment 2	26%	Low
Segment 3	64%	High
Segment 4	81%	Severe
Segment 5	43%	Moderate

13. Arrival-Time-Aware Prediction

FLOODRIX considers the estimated time at which the vehicle will reach each road segment. This allows the system to use the rainfall conditions relevant to the expected arrival period rather than treating flood risk as a static property.

Departure: 10:00 AM

```
Segment 1 → ETA 10:45 → Risk 12%
Segment 2 → ETA 11:30 → Risk 22%
Segment 3 → ETA 12:40 → Risk 67%
Segment 4 → ETA 01:20 → Risk 79%
```

The goal is to answer: **What is the expected flood risk at this location around the time the vehicle reaches it?**

14. Route Risk Engine

Flood Probability
+
Road Closure
+
ETA
+
Cargo
+
Delivery Deadline
↓
Route Risk Evaluation

The ML model produces the fundamental flood probability. The logistics engine then uses that output alongside transportation constraints. The ML model does not directly decide which route the truck takes.

15. Logistics Parameters

Category	Parameters
Shipment	Cargo type, cargo priority, delivery deadline
Vehicle	Vehicle type, vehicle capacity, current location
Route	Distance, travel time, road segments, predicted flood probability
Disruption	Road closures, waterlogging reports, verified disruptions

16. Cargo-Aware Decision Making

Cargo type belongs to the logistics decision layer rather than the flood-prediction model. It determines how much delay or route risk is acceptable.

Cargo	Typical Sensitivity
Perishable	High sensitivity to delay
FMCG	Medium sensitivity
Non-perishable	Lower sensitivity

17. Road Closure Layer

Road closures are treated as an external logistics constraint. The system records road segment, location, closure status, timestamp, source, severity, and expiry.

Road Closure
↓
Route invalidated
↓
Alternative routes generated
↓
Flood prediction applied
↓
New route recommendation

18. Route Recommendation

The final recommendation should be understandable to the operator.

Route	Distance	ETA	Flood Exposure	Decision
Route A	165 km	4h 05m	High	Alternative
Route B	178 km	4h 32m	Low	Recommended

Example explanation: **Route B is recommended because it has lower predicted flood exposure despite being longer.**

19. Live Monitoring

```

Vehicle GPS
↓
Current location
↓
Upcoming route segments
↓
Updated rainfall
↓
Updated flood prediction
↓
Road closure updates
↓
Risk recalculation

```

- Continue
- Warning
- Reroute recommendation
- Route blocked
- ETA update

20. Alert System

Alert	Trigger	Action
Flood Risk Alert	High predicted probability ahead	Warn operator
Increasing Risk	Significant risk change	Recalculate route
Road Closure	Verified closure	Generate alternatives
Reroute	Safer alternative available	Recommend alternative
ETA	Significant delay	Notify operator

21. Monsoon Simulation

The simulation mode allows operators to test what-if rainfall scenarios. The simulator should pass modified rainfall parameters through the same prediction and route evaluation pipeline rather than returning hardcoded outputs.

```

Current rainfall
↓
Prediction
↓
Flood probability

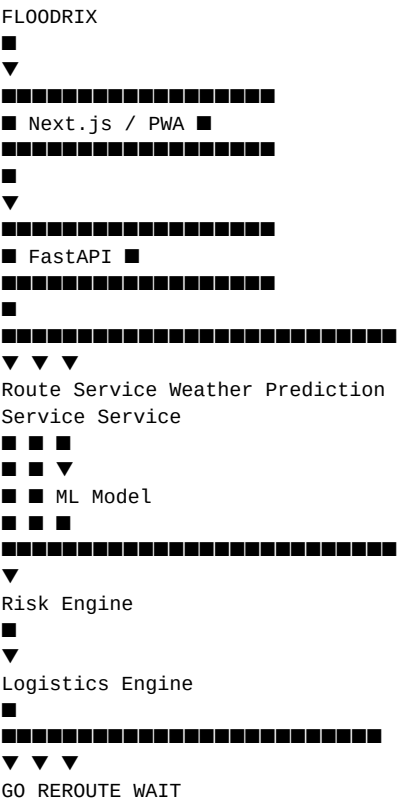
Simulated rainfall change
↓
Prediction
↓
New flood probability
↓

```

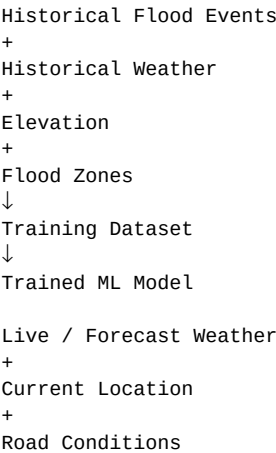
22. Main User Interface

Screen	Purpose
Command Center	Active trips, high-risk areas, current alerts, weather and route status
Plan Shipment	Origin, destination, cargo, vehicle, departure and deadline
Route Intelligence	Map, routes, flood probability, risk segments, ETA and recommendation
Live Trip	Vehicle location, upcoming risk, alerts, ETA and rerouting
Simulation	Rainfall scenario, predicted flood probability, route changes and impact

23. System Architecture



24. Data Architecture



+
Closures
↓
Prediction + Logistics Layer

25. Database Schema

Table	Core Fields
flood_training_data	id, latitude, longitude, rainfall_1h, rainfall_6h, rainfall_24h, elevation, historical_flood_frequency, flood_zone, flood_o
flood_predictions	id, latitude, longitude, predicted_at, flood_probability, risk_level, model_version
trips	id, origin, destination, cargo_type, vehicle_type, departure_time, deadline, status
route_segments	id, route_id, geometry, arrival_time, flood_probability, risk_level
closures	id, road_segment_id, location, severity, source, reported_at, expires_at, status

26. API Requirements

Flood Prediction

POST /api/flood/predict

Request fields: latitude, longitude, rainfall_1h, rainfall_6h, rainfall_24h, elevation, historical_flood_frequency, flood_zone.

Response: flood_probability and risk_level.

Route Planning

POST /api/routes/plan

Request: origin, destination, cargo_type, vehicle_type, departure_time, deadline.

Response: recommended route, route alternatives, flood exposure, ETA, alerts and recommendation reason.

Other Core APIs

GET /api/flood/events
GET /api/routes/{route_id}/risk
POST /api/closures
GET /api/closures
POST /api/simulate
WS /ws/alerts

27. ML Model Requirements

The initial implementation should evaluate classification models such as Logistic Regression, Random Forest, and XGBoost. The final model should be selected based on validation performance rather than assumed beforehand.

Evaluation metrics:

- Precision
- Recall
- F1 Score
- ROC-AUC
- Confusion Matrix

Recall is particularly important because a false negative can cause the system to underestimate a dangerous location.

28. Data Leakage Prevention

The model must not use information that would only become available after a flood occurred when making a future prediction.

Historical actual flood occurrence is the target label, not a future input feature.

```
TRAINING
Features → Rainfall + Location + Elevation + Historical information
Target → Flood_occurred

PREDICTION
Current/forecast features
↓
Trained model
↓
Flood probability
```

29. Success Criteria

ML

```
Precision: TBD
Recall: TBD
F1: TBD
ROC-AUC: TBD
```

Metrics will be populated only after the model is trained and evaluated on unseen data.

Product

- Generate a route.
- Calculate flood probability for route segments.
- Identify high-risk segments.
- Compare alternative routes.
- Incorporate road closures.
- Produce an ETA.
- Recommend a route.
- Update risk when conditions change.

30. MVP Scope

Must Have

- Historical flood dataset
- Historical rainfall dataset
- Training pipeline
- Flood prediction model
- Forecast rainfall integration
- Route generation
- Segment-level prediction
- Route risk evaluation
- Road closure layer

- Route recommendation
- Map visualization

Should Have

- Live vehicle tracking
- Alerts
- Cargo-aware routing
- Simulation

Future

- Advanced hydrological variables
- River-level prediction
- Landslide prediction
- Satellite imagery
- Statewide deployment
- Multimodal transport
- Cold-storage recommendations
- WhatsApp/SMS notifications
- Advanced deep-learning models

31. Key Differentiator

Traditional navigation asks:

“What is the fastest route?”

FLOODRIX asks:

“What is the probability of flooding along my route, and which route gives me the best balance between travel time and flood risk?”

```

Historical Flood Events
+
Rainfall Conditions
+
Geographic Characteristics
↓
Machine Learning
↓
Flood Probability
↓
Route-Level Risk
↓
Logistics Decision

```

32. End-to-End Demo Scenario

The official challenge provides a vegetable truck travelling from Palakkad to Kochi and encountering heavy flooding as an example scenario.

```

Palakkad
↓
Create shipment
↓
Vegetable cargo

```

↓
Generate routes
↓
Sample route segments
↓
Obtain rainfall conditions
↓
Generate model features
↓
Predict flood probability
↓
Identify high-risk segments
↓
Check road closures
↓
Compare alternative routes
↓
Select safest practical route
↓
Monitor during journey
↓
Recalculate when conditions change
↓
Reroute if required

33. Final Product Definition

FLOODRIX is a machine-learning-powered flood prediction and logistics intelligence platform that learns from historical flood occurrences and associated environmental conditions to estimate flood probability at locations along a vehicle's route. This prediction is combined with route, cargo, ETA, delivery deadline, and road-closure information to provide predictive and risk-aware logistics decisions.

34. Technology Stack

Layer	Technology
Frontend	Next.js, React, TypeScript, Tailwind CSS, PWA
Backend	Python, FastAPI
Machine Learning	Python, Scikit-learn, XGBoost / Random Forest
Database	PostgreSQL, PostGIS
Weather	Open-Meteo, IMD
Flood Data	KSDMA
Routing	Google Routes API / OpenStreetMap-based routing

35. Risks & Limitations

- Historical flood data may have incomplete spatial or temporal coverage.
- Weather forecasts contain uncertainty.
- Flood occurrence labels may vary in quality between datasets.
- Road-closure information may not always be complete or verified.
- Model performance depends on training-data quality and geographic coverage.
- The system provides decision support and does not replace official disaster-management warnings.

36. Roadmap

Phase	Deliverable
Phase 1	Historical flood and weather dataset preparation
Phase 2	Feature engineering and model training
Phase 3	Prediction and route-risk APIs
Phase 4	Map-based frontend
Phase 5	Weather, closures and live integration
Phase 6	Simulation, testing and hackathon demo

37. Conclusion

FLOODRIX is designed to move flood-aware logistics from static historical mapping toward predictive, route-specific decision support. Its core ML model learns from historical flood occurrence and associated rainfall, geographic and flood-zone features. The resulting flood probability is then connected to real logistics constraints so that operators can understand risk, compare routes, and act before a disruption becomes a delivery failure.

Predict. Protect. Deliver.